

BHARATHIAR UNIVERSITY

State University (Re-Accredited with “A⁺⁺” Grade by NAAC)
COIMBATORE - 641046



MASTER OF COMPUTER APPLICATIONS

(JULY 2024 BATCH)

SEMESTER – II

LAB PRACTICAL V:

Web Application Development and Hosting

SCHOOL OF DISTANCE EDUCATION

CENTRE FOR DISTANCE AND ONLINE EDUCATION

OPEN AND DISTANCE LEARNING (ODL)

DECEMBER 2025



SCHOOL OF DISTANCE EDUCATION
CENTRE FOR DISTANCE AND ONLINE EDUCATION
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BHARATHIAR UNIVERSITY

CERTIFICATE

Enrolment No. :
Name of the Course : Master of Computer Applications
Subject : Web Application Development and Hosting

Certified that this is a Bonafide record of practical work done by

Mr. / Ms. _____

at the Computer Centre – I / II, School of Distance Education, Bharathiar University,
Coimbatore – 46 during Academic / Calendar Year 20 - 20

SIGNATURE OF FACULTY

SIGNATURE OF COORDINATOR

Submitted for the Practical Examination held on _____
at the Computer Centre – I / II, School of Distance Education, Bharathiar University,
Coimbatore – 641 046.

INTERNAL EXAMINER

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EXNO:01	Develop a website for your college using advanced HTML tags
DATE:	

Aim:

To design and develop a static web page using HTML and CSS for “**Bharathiyar University**” that contains header, navigation menu, about section, courses table, contact form and footer

Algorithm:

Step 1: Start the program by declaring `<!DOCTYPE html>` and open the `<html>` tag.

Step 2: Add the `<head>` section and include meta tags, page title and internal CSS styling.

Step 3: Open the `<body>` tag to begin the visible part of the webpage.

Step 4: Insert the `<header>` tag and add the university name and slogan text.

Step 5: Create a `<nav>` navigation bar and add links such as About, Courses and Contact.

Step 6: Add an `<section>` for the About content and include `<details>` and `<summary>` to show additional information.

Step 7: Add another `<section>` for Courses and insert a table containing course name, duration and type.

Step 8: Create a Contact `<section>` and add a form with input fields like name, email, message and a submit button.

Step 9: Insert the `<footer>` tag for copyright details and close the `<body>` and `<html>` tags to complete the web page.

Source code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Bharathiyar University</title>
  <style>
    body {
```

```
    font-family: Arial, sans-serif;
    margin: 0;
    background: #f4f6f9;
}
header {
    background: #002147;
    color: white;
    padding: 20px;
    text-align: center;
}
nav {
    background: #064579;
    padding: 10px;
}
nav a {
    color: white;
    margin: 15px;
    text-decoration: none;
    font-weight: bold;
}
section {
    padding: 20px;
}
article {
    background: white;
    padding: 20px;
    margin-bottom: 20px;
    border-radius: 5px;
```

```

    }
    footer {
        background: #002147;
        color: white;
        text-align: center;
        padding: 10px;
    }
    table {
        width: 100%;
        border-collapse: collapse;
        background: white;
    }
    th, td {
        padding: 12px;
        border: 1px solid #ccc;
    }
    iframe {
        width: 100%;
        height: 300px;
        border: none;
    }
</style>
</head>
<body>
<header>
    <h1> Bharathiyar University </h1>
    <p>"Educate, Empower, Enlighten"</p>
</header>

```

```

<nav>
  <a href="#about">About</a>
  <a href="#courses">Courses</a>
  <a href="#contact">Contact</a>
</nav>
<section id="about">
  <article>
    <h2>About Our University</h2>
    <p>Bharathiyar University is one of the leading educational institutions offering quality
    education in arts, science, engineering and advanced research.</p>
    <!-- Advanced HTML Tag: <details> -->
    <details>
      <summary>Read More About Our History</summary>
      <p>Founded in 1982, the university is named after the great Tamil poet Subramania
      Bharathiyar.</p>
    </details>
  </article>
</section>
<section id="courses">
  <article>
    <h2>Available Courses</h2>
    <!-- Advanced HTML Tag: Table -->
    <table>
      <tr>
        <th>Course Name</th>
        <th>Duration</th>
        <th>Type</th>
      </tr>
      <tr>

```

```

        <td>B.Sc Computer Science</td>
        <td>3 Years</td>
        <td>Full-time</td>
    </tr>
    <tr>
        <td>BBA</td>
        <td>3 Years</td>
        <td>Full-time</td>
    </tr>
    <tr>
        <td>MCA</td>
        <td>2 Years</td>
        <td>Post Graduate</td>
    </tr>
</table>
</article>
</section>
<section id="contact">
    <article>
        <h2>Contact Us</h2>
        <!-- Advanced HTML Tag: HTML5 Form -->
        <form>
            <label>Name: </label>
            <input type="text" placeholder="Enter your name" required><br><br>
            <label>Email: </label>
            <input type="email" placeholder="Enter your email" required><br><br>
            <label>Message: </label><br>
            <textarea rows="5" cols="40"></textarea><br><br>

```



```

        <button type="submit">Submit</button>

    </form>

</article>

</section>

<footer>

    <p>© 2025 Bharathiyar University. All Rights Reserved.</p>

</footer>

</body>

</html>

```

Output:

Bharathiyar University

"Educate, Empower, Enlighten"

[About](#)
[Courses](#)
[Contact](#)

About Our University

Bharathiyar University is one of the leading educational institutions offering quality education in arts, science, engineering and advanced research.

► [Read More About Our History](#)

Available Courses

Course Name	Duration	Type
B.Sc Computer Science	3 Years	Full-time
BBA	3 Years	Full-time
MCA	2 Years	Post Graduate

Contact Us

Name:

Email:

Message:

© 2025 Bharathiyar University. All Rights Reserved.

Result:

The output has been executed and verified successfully.

EXNO:02	Paragraph with country names as hyperlinks (world.html)
DATE:	

Aim:

To develop a web page containing a paragraph with country names as hyperlinks, where selecting each hyperlink opens a separate webpage displaying information about that country.

Algorithm:

Step 1: Start the program by creating a new HTML file named world.html.

Step 2: Declare `<!DOCTYPE html>` and open `<html>` and `<head>` tags.

Step 3: Add a suitable title to the webpage in the `<title>` tag.

Step 4: Open the `<body>` tag to display webpage content.

Step 5: Create a paragraph in the body section.

Step 6: Inside the paragraph, insert hyperlinks using the `<a>` tag for each country (India, Japan, US).

Step 7: Create separate HTML files named India.html, Japan.html, and US.html.

Step 8: Add information related to each country inside its own web page.

Step 9: Execute the program in a browser and verify that clicking each country link opens the corresponding webpage.

Source code:

world.html

```
<!DOCTYPE html>

<html>

<head>  <title>World Countries</title> </head>

<body>

    <h2>World Page</h2>

    <p>
```

Click on a country to know more:

`<a href="India.html">India</a>,`

`<a href="Japan.html">Japan</a>,`

`<a href="US.html">United States</a>`

`</p>`

`</body>`

`</html>`

India.html

`<!DOCTYPE html>`

`<html>`

`<head> <title>India</title></head>`

`<body>`

`<h2>India</h2>`

`<p>India is a country in South Asia known for its rich culture, diversity, and history.</p>`

`<a href="world.html">Back to World</a>`

`</body> </html>`

Japan.html

`<!DOCTYPE html>`

`<html>`

`<head> <title>Japan</title></head>`

```
<body>
```

```
<h2>Japan</h2>
```

```
<p>Japan is an East Asian island nation famous for its technology, tradition, and cuisine.</p>
```

```
<a href="world.html">Back to World</a>
```

```
</body>
```

```
</html>
```

US.html

```
<!DOCTYPE html>
```

```
<html>
```

```
<head> <title>United States</title></head>
```

```
<body>
```

```
<h2>United States</h2>
```

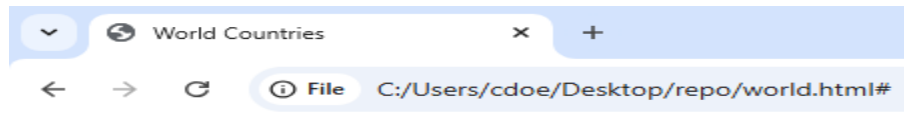
```
<p>The United States is a country in North America known for its economy, innovation, and global influence.</p>
```

```
<a href="world.html">Back to World</a>
```

```
</body>
```

```
</html>
```

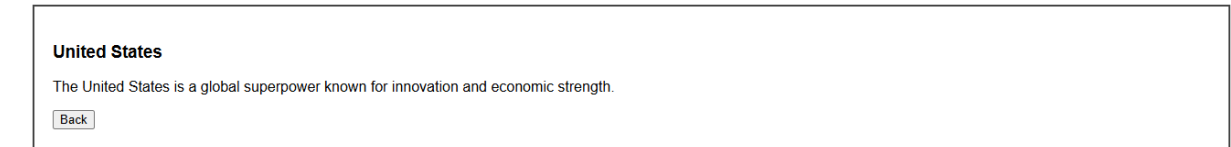
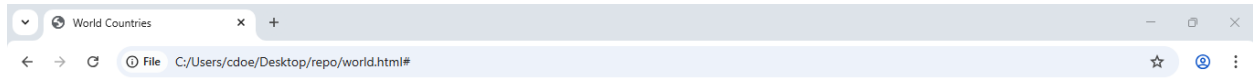
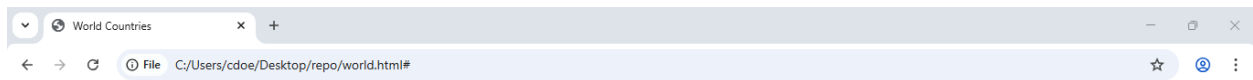
Output:



World Page

Click a country to view information:

[India](#) | [Japan](#) | [United States](#)



Result:

The above program has been executed and verified successfully.

EXNO:03	Displaying text using Lists and Tables in HTML
DATE:	

Aim:

To develop an HTML program to display text information using different types of lists and tables.

Algorithm:

Step 1: Start the HTML program using the `<!DOCTYPE html>` declaration and open the `<html>` tag.

Step 2: Add the `<head>` section and include a suitable `<title>`.

Step 3: Open the `<body>` tag to begin displaying content on the webpage.

Step 4: Insert a heading to describe the purpose of the page.

Step 5: Create an unordered list (`<ul>`) and add list items to display information using bullet points.

Step 6: Create an ordered list (`<ol>`) and add list items to display information using numbers.

Step 7: Insert a table using the `<table>` tag and define borders and structure to organize content in rows and columns.

Step 8: Add table headers (`<th>`) and rows (`<tr>`) with table data (`<td>`) to present the information in tabular format.

Step 9: Close the `<body>` and `<html>` tags and finally execute the program in a web browser.

Source code:

```
<!DOCTYPE html>

<html>

<head>  <title>Lists and Tables Example</title></head>

<body>

<h2>Displaying Text Using Lists</h2>
```

<h3>Unordered List:</h3>

HTML

CSS

JavaScript

<h3>Ordered List:</h3>

Red

Green

Blue

<h2>Displaying Text Using Table</h2>

<table border="1" cellpadding="8">

<tr>

<th>Name</th>

<th>Department</th>

<th>Year</th>

</tr>

<tr>

<td>Asha</td>

<td>Computer Science</td>

<td>2025</td>

```
</tr>
```

```
<tr>
```

```
<td>Ravi</td>
```

```
<td>Information Technology</td>
```

```
<td>2024</td>
```

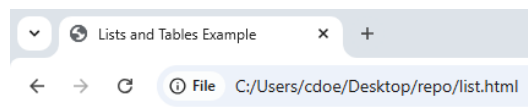
```
</tr>
```

```
</table>
```

```
</body>
```

```
</html>
```

Output:



Displaying Text Using Lists

Unordered List:

- HTML
- CSS
- JavaScript

Ordered List:

1. Red
2. Green
3. Blue

Displaying Text Using Table

Name	Department	Year
Asha	Computer Science	2025
Ravi	Information Technology	2024

Result:

The above program has been executed and verified successfully.

EXNO:04	Complete Web Page about a Hospital using Frames and Framesets
DATE:	

Aim:

To design and develop a complete web page of a hospital using HTML frames and framesets so that different sections of a website load within the same browser window.

Algorithm:

Step 1: Start the HTML program using the `<!DOCTYPE html>` declaration and open the `<html>` tag.

Step 2: Use the `<frameset>` tag to divide the browser window into multiple frames.

Step 3: Create the top frame to display the hospital heading or logo.

Step 4: Create the left frame to display a navigation menu with links.

Step 5: Create the remaining frame to display the main content area.

Step 6: Add hyperlinks in the menu so that clicking each link loads a corresponding page inside the main frame.

Step 7: Insert `<frame>` tags to embed separate HTML files inside the frames.

Step 8: Close the `<frameset>` and `<html>` tags properly.

Step 9: Execute the file in a browser and verify that each menu option loads inside the content frame without refreshing the full page.

Source code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>  <title>Hospital Website</title>  <style>
```

```
    body { margin: 0; font-family: Arial; }
```

```
    #top { background: #005f85; color: white; text-align: center; padding: 20px; }
```

```

#left { background: #e5f7ff; width: 25%; height: 600px; float: left; padding: 15px; }

#main { width: 73%; height: 600px; float: right; border-left: 2px solid #ccc; }

a { display: block; margin-bottom: 15px; font-size: 18px; text-decoration: none; color:
#005f85; }

</style>

<script>

function loadContent(page) {

    document.getElementById("content").src = page;

}

</script> </head>

<body>

<!-- Top Frame -->

<div id="top">

    <h1>City Care Hospital</h1>

    <p>Your Health • Our Priority</p>

</div>

<div id="left">

    <a href="#" onclick="loadContent('about')">About Us</a>

    <a href="#" onclick="loadContent('services')">Services</a>

```

`<a href="#" onclick="loadContent('contact')">Contact</a>`

`</div>`

`<div id="main">`

`<iframe id="content" style="width:100%; height:100%; border:none;"></iframe>`

`</div>`

`<template id="about">`

`<h2>About Us</h2>`

`<p>City Care Hospital is a multi-specialty healthcare centre providing 24x7 medical services.</p>`

`</template>`

`<template id="services">`

`<h2>Our Services</h2>`

`<ul>`

`<li>Emergency & ICU</li>`

`<li>Cardiology</li>`

`<li>Orthopedics</li>`

`<li>Maternity Care</li>`

`<li>Pharmacy & Diagnostics</li>`

`</ul> </template>`

```
<template id="contact">
```

```
<h2>Contact Information</h2>
```

```
<p>Phone: +91 98765 43210</p>
```

```
<p>Email: citycare@hospital.com</p>
```

```
<p>Location: MG Road, Coimbatore</p>
```

```
</template>
```

```
<script>
```

```
function loadContent(page) {
```

```
    const iframe = document.getElementById("content");
```

```
    const temp = document.getElementById(page);
```

```
    iframe.srcdoc = temp.innerHTML; // loads hidden page inside iframe
```

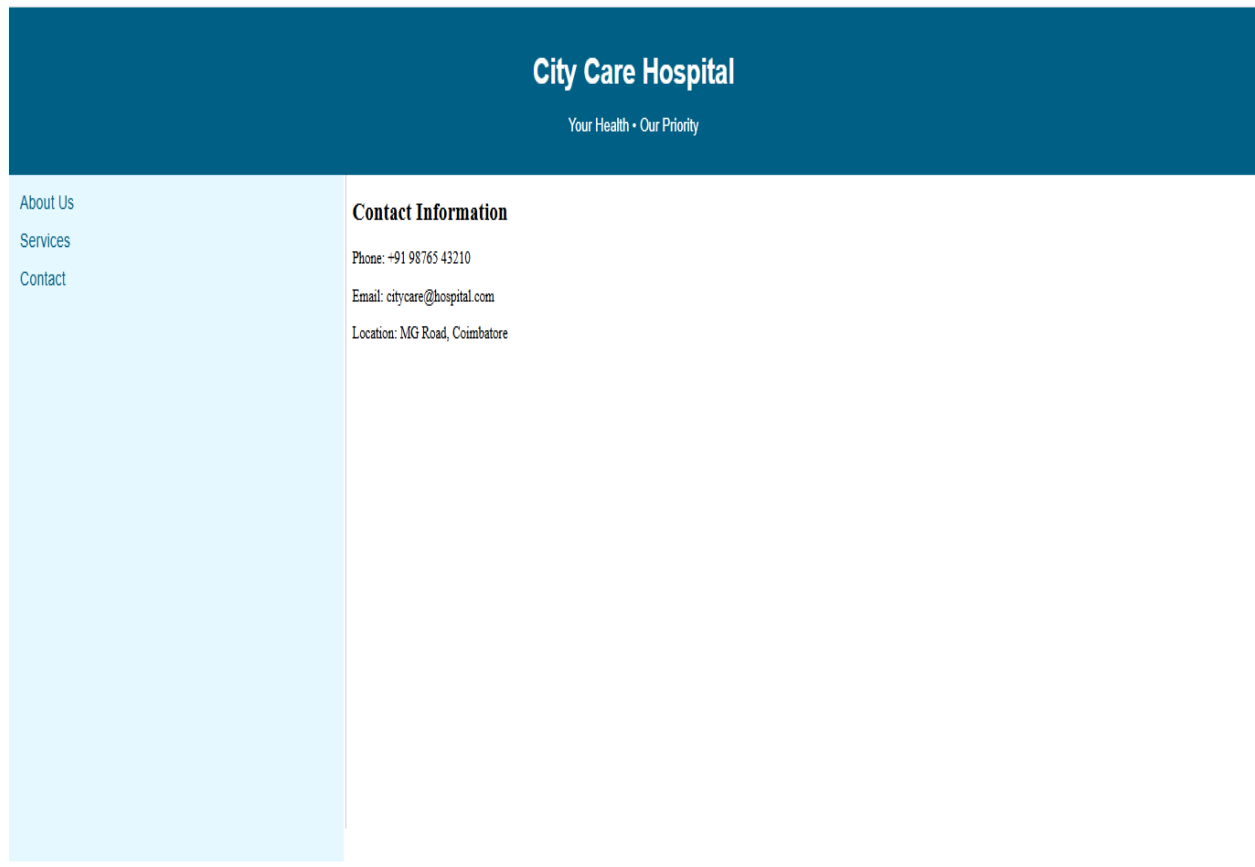
```
}
```

```
</script>
```

```
</body>
```

```
</html>
```

Output:



Result:

The above program has been executed and verified successfully.

EXNO:05	HTML document to print your Bio-Data in a neat format
DATE:	

Aim:

To design a web page in HTML that collects personal bio-data information from the user through a form and displays the submitted information on the same page using JavaScript.

Algorithm:

Step 1: Start the program.

Step 2: Create an HTML document structure using `<!DOCTYPE html>`, `<html>`, `<head>`, and `<body>` tags.

Step 3: Design the layout of the form using HTML controls such as `textbox`, `textarea`, `radio` button, `date`, etc.

Step 4: Add labels and input fields to collect details like name, DOB, gender, email, mobile number, address, city, state, qualifications, and hobbies.

Step 5: Insert a **Submit** button.

Step 6: Write a **JavaScript function (showData())** that reads the values entered by the user using `document.getElementById()` and `document.querySelector()`.

Step 7: Display the collected data inside a `<div>` section using `innerHTML`.

Step 8: Link the JavaScript function to the Submit button using `onclick`.

Step 9: Save the file and run it in a web browser to verify the output.

Source code:

```
<!DOCTYPE html>

<html>

<head>  <title>Personal Bio-Data Form</title>  <style>

    body {

        font-family: "Segoe UI", sans-serif;
```

```
background: #f0f5ff;

margin: 0;

padding: 0;

}

.container {

width: 450px;

background: white;

padding: 25px;

margin: 40px auto;

border-radius: 12px;

box-shadow: 0 0 15px rgba(0,0,0,0.2);

}

h2 {

text-align: center;

color: #003d99;

margin-bottom: 20px;

}

label {

font-weight: bold;
```

```
display: block;

margin-top: 10px;

}

input, textarea, button, select {

width: 100%;

padding: 8px;

margin-top: 5px;

border-radius: 6px;

border: 1px solid #aaa;

}

button {

background: #003d99;

color: white;

border: none;

margin-top: 15px;

font-size: 16px;

cursor: pointer;

padding: 10px;

}
```



```

button:hover {

    background: #002b73;

}

#result {

    margin-top: 25px;

    padding: 20px;

    background: #e6f0ff;

    border-left: 4px solid #003d99;

    border-radius: 8px;

}

p {

    margin: 6px 0;

}

</style>  <script>

function showData() {

    document.getElementById("result").innerHTML = `

        <h3>Your Bio-Data</h3>

        <p><b>Full Name:</b> ${document.getElementById("name").value}</p>

```

```

    <p><b>Date of Birth:</b> ${document.getElementById("dob").value}</p>

    <p><b>Gender:</b>
    ${document.querySelector('input[name="gender"]:checked').value}</p>

    <p><b>Email:</b> ${document.getElementById("email").value}</p>

    <p><b>Phone Number:</b> ${document.getElementById("phone").value}</p>

    <p><b>Address:</b> ${document.getElementById("address").value}</p>

    <p><b>City:</b> ${document.getElementById("city").value}</p>

    <p><b>State:</b> ${document.getElementById("state").value}</p>

    <p><b>Qualifications:</b> ${document.getElementById("qual").value}</p>

    <p><b>Hobbies:</b> ${document.getElementById("hobbies").value}</p> `;

} </script> </head> <body>

```

```

<div class="container">

```

```

    <h2>Personal Bio-Data Form</h2>

```

```

    <label>Full Name:</label>

```

```

    <input type="text" id="name">

```

```

    <label>Date of Birth:</label>

```

```

    <input type="date" id="dob">

```

<label>Gender:</label>

<input type="radio" name="gender" value="Male"> Male

<input type="radio" name="gender" value="Female"> Female

<input type="radio" name="gender" value="Other"> Other

<label>Email:</label>

<input type="email" id="email">

<label>Phone Number:</label>

<input type="text" id="phone">

<label>Address:</label>

<textarea id="address" rows="3"></textarea>

<label>City:</label>

<input type="text" id="city">

<label>State:</label>

<input type="text" id="state">

<label>Qualifications:</label>

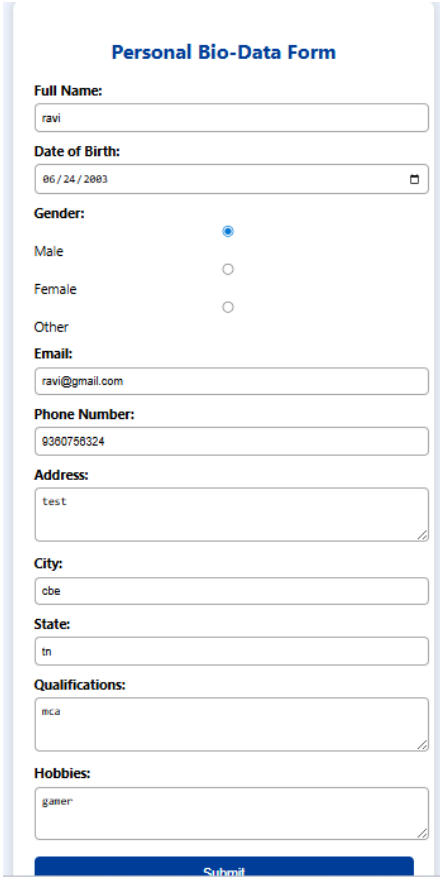
<textarea id="qual" rows="3"></textarea> <label>Hobbies:</label>

<textarea id="hobbies" rows="3"></textarea>

<button onclick="showData()">Submit</button>

```
<div id="result"></div> </div> </body> </html>
```

Output:



Personal Bio-Data Form

Full Name:
ravi

Date of Birth:
06 / 24 / 2003

Gender:
☒ Male
☐ Female
☐ Other

Email:
ravi@gmail.com

Phone Number:
9360756324

Address:
test

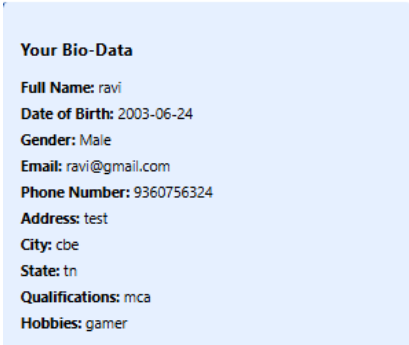
City:
cbe

State:
tn

Qualifications:
mca

Hobbies:
gamer

Submit



Your Bio-Data

Full Name: ravi
Date of Birth: 2003-06-24
Gender: Male
Email: ravi@gmail.com
Phone Number: 9360756324
Address: test
City: cbe
State: tn
Qualifications: mca
Hobbies: gamer

Result:

The above program has been executed and verified successfully.

EXNO:06	Registration Form with JavaScript validation
DATE:	

Aim:

To develop a **Registration Form web page** using HTML and JavaScript that collects user details and validates the input fields before submitting the form, ensuring that all required information is entered correctly.

Algorithm:

Step 1: Start the program.

Step 2: Create an HTML document with a registration form containing fields such as Name, Email, Password, Confirm Password, Phone Number, and other required details.

Step 3: Add a Submit button to the form.

Step 4: Write a JavaScript validation function to check:

- Whether mandatory fields are filled
- Whether email format is valid
- Whether password and confirm password match
- Whether phone number is correct length/numbers

Step 5: Use `document.getElementById()` to access the values entered by the user.

Step 6: Display error messages using `alert()` or `<span>` for invalid input.

Step 7: If all fields are valid, allow form submission; otherwise prevent submission.

Step 8: Link the validation function to the Submit button using `onsubmit` or `onclick`.

Step 9: Save the file as `registration.html` and run it in a browser to verify the output.

Source code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head> <title>Registration Form</title> <style>
```

```
body {
```

```
font-family: Arial, sans-serif;

background: #f2f2f2;

margin: 0;

padding: 0;

}
```

```
.container {

width: 350px;

background: white;

padding: 20px;

margin: 50px auto;

border-radius: 10px;

box-shadow: 0 0 10px #aaa;

}
```

```
h2 {

text-align: center;

color: #333;

}
```

```
label {

display: block;
```

```
margin-top: 10px;

font-weight: bold;
}

input {

width: 100%;

padding: 7px;

margin-top: 5px;

border-radius: 5px;

border: 1px solid #ccc;

box-sizing: border-box;
}

button {

width: 100%;

padding: 10px;

margin-top: 15px;

background: #0275d8;

color: white;

border: none;

border-radius: 5px;
```

```
        font-size: 16px;

        cursor: pointer;
    }

    button:hover {

        background: #025aa5;

    }

    #notification {

        margin-top: 15px;

        padding: 10px;

        background: #d4edda;

        color: #155724;

        border-radius: 5px;

        display: none;

        text-align: center;

    }

</style>

<script>

    function validateForm(event) {

        event.preventDefault(); // prevent default form submission
```



```

var name = document.getElementById("name").value.trim();

var place = document.getElementById("place").value.trim();

var email = document.getElementById("email").value.trim();

var phone = document.getElementById("phone").value.trim();

if(name === "") { alert("Enter Name"); return; }

if(place === "") { alert("Enter College/Place"); return; }

if(email === "") { alert("Enter Email"); return; }

var emailPattern = /^[^ ]+@[^ ]+\.[a-z]{2,3}$/;

if(!emailPattern.test(email)) { alert("Invalid Email"); return; }

if(phone === "" || phone.length != 10 || isNaN(phone)) { alert("Invalid Phone"); return; }

// Display notification

var note = document.getElementById("notification");

note.innerHTML = `Thank you ${name}! Your registration is successful.`;

note.style.display = "block";

// Optional: Clear the form

document.getElementById("regForm").reset();

}

</script></head><body>

```

```
<div class="container">

  <h2>Registration Form</h2>

  <form id="regForm" onsubmit="validateForm(event)">

    <label>Name:</label>

    <input type="text" id="name" placeholder="Enter your full name">

    <label>College/Place:</label>

    <input type="text" id="place" placeholder="Enter your college/place">

    <label>Email:</label>

    <input type="email" id="email" placeholder="Enter your email">

    <label>Phone Number:</label>

    <input type="text" id="phone" placeholder="Enter 10-digit phone number">

    <button type="submit">Submit</button>

  </form>

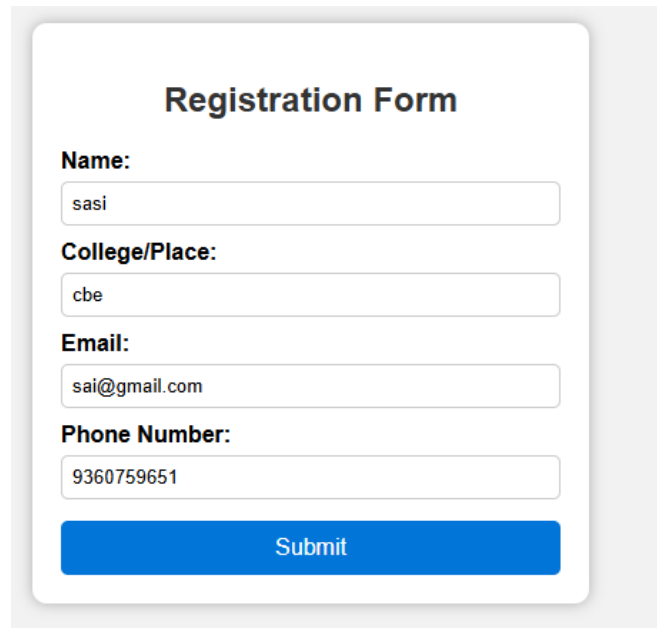
  <div id="notification"></div>

</div>

</body>

</html>
```

Output:



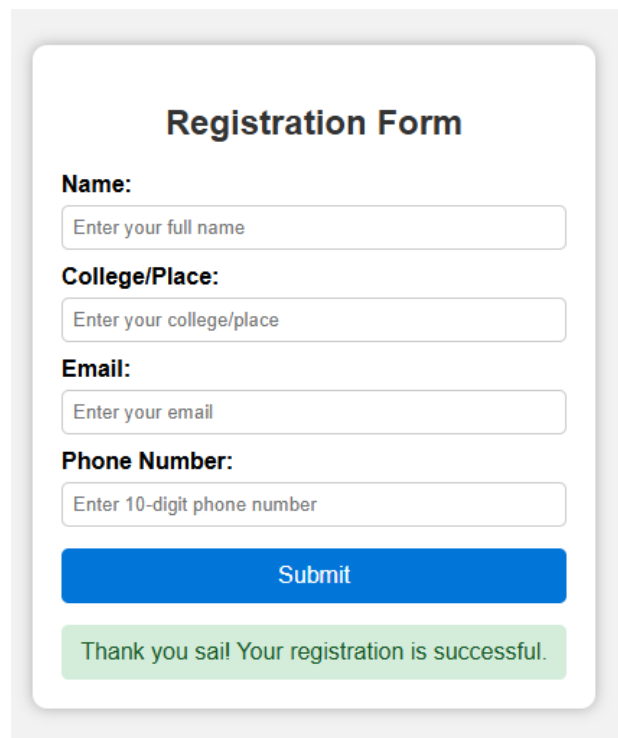
Registration Form

Name:

College/Place:

Email:

Phone Number:



Registration Form

Name:

College/Place:

Email:

Phone Number:

Thank you sai! Your registration is successful.

Result:

The above program has been executed and verified successfully.

EXNO:07	Display customer details using XML, XSL and validate with DTD/XSD
DATE:	

Aim:

To develop an XML document for storing customer details and transform it into a formatted webpage using XSLT. The XML data should be validated using **DTD/XSD** to ensure correctness before displaying it through JavaScript.

Algorithm:

Step 1: Start the program.

Step 2: Create an XML file to store customer details.

Step 3: Add a DTD/XSD file to validate XML structure and data types.

Step 4: Create an XSL file to format the XML data into an HTML table.

Step 5: In JavaScript, load both XML and XSL files using `XMLHttpRequest`.

Step 6: Apply the XSL transformation using `XSLTProcessor`.

Step 7: Display the transformed data inside the webpage.

Step 8: Ensure the XML validates successfully with DTD/XSD.

Step 9: Execute the project in a browser and view the formatted output.

Source code:

customer.xml

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<customers>
```

```
  <customer id="C001">
```

```
    <name>John Doe</name>
```

```
    <email>john@example.com</email>
```

```
    <phone>9876543210</phone>
```

</customer>

<customer id="C002">

<name>Jane Smith</name>

<email>jane@example.com</email>

<phone>9123456780</phone>

</customer>

<customer id="C003">

<name>Michael Brown</name>

<email>michael@example.com</email>

<phone>9988776655</phone>

</customer>

</customers>

customer.xsl

<?xml version="1.0" encoding="UTF-8"?>

<xsl:stylesheet version="1.0"

xmlns:xsl="http://www.w3.org/1999/XSL/Transform">

<xsl:template match="/">

<html>

<head>

```
<style>

  table {

    border-collapse: collapse;

    width: 60%;

  }

  th, td {

    border: 1px solid #000;

    padding: 8px;

  }

  th {

    background: #eee;

  }

</style>

</head>

<body>

  <table>

    <tr>

      <th>ID</th><th>Name</th><th>Email</th><th>Phone</th>

    </tr>
```

```

        <xsl:for-each select="customers/customer">

        <tr>

            <td><xsl:value-of select="@id"/></td>

            <td><xsl:value-of select="name"/></td>

            <td><xsl:value-of select="email"/></td>

            <td><xsl:value-of select="phone"/></td>

        </tr>

        </xsl:for-each>

    </table>

</body>

</html>

</xsl:template> </xsl:stylesheet>
index.html

<!DOCTYPE html>

<html>

<head>

    <meta charset="UTF-8">

    <title>Customer Details</title>

```

```
</head>
```

```
<body>
```

```
<h1>Customer Details (XSLT - Works in Edge)</h1>
```

```
<div id="output"></div>
```

```
<script>
```

```
// Load XML & XSL, then transform
```

```
async function loadXML(url) {
```

```
    const response = await fetch(url);
```

```
    const text = await response.text();
```

```
    return new window.DOMParser().parseFromString(text, "text/xml");
```

```
}
```

```
async function transform() {
```

```
    const xml = await loadXML("customer.xml");
```

```
    const xsl = await loadXML("customer.xsl");
```

```
    const processor = new XSLTProcessor();
```

```
    processor.importStylesheet(xsl);
```

```
    const result = processor.transformToFragment(xml, document);
```

```
    document.getElementById("output").appendChild(result);
```

```
}
```



```
transform();
```

```
</script></body></html>
```

Output:



Customer Details (XSLT - Works in Edge)

ID	Name	Email	Phone
C001	John Doe	john@example.com	9876543210
C002	Jane Smith	jane@example.com	9123456780
C003	Michael Brown	michael@example.com	9988776655

Result:

The above program has been executed and verified successfully.

EXNO:08	Student personal details in XML format
DATE:	

Aim:

To store student personal details in XML format and write a JavaScript program using the **xml2js** parser to read the XML file and display selected details such as student name and course..

Algorithm:

Step 1: Start the program.

Step 2: Create an XML file named **stu.xml** containing student personal details.

Step 3: Import the required Node.js modules **fs** (File System) and **xml2js** to read and parse XML.

Step 4: Read the XML file contents using the **fs.readFileSync()** function.

Step 5: Create an XML parser object using **new xml2js.Parser()**.

Step 6: Convert the XML data into JSON format using the **parser.parseString()** method.

Step 7: Retrieve the **student** array from the parsed XML result.

Step 8: Loop through each student object and display the required details (Name and Course) on the console.

Step 9: Stop / End the program.

Source code:

stu.xml

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<students>
```

```
  <student>
```

```
    <id>1</id>
```

```
    <name>John Doe</name>
```

<dob>2000-05-10</dob>

<gender>Male</gender>

<email>john@example.com</email>

<phone>1234567890</phone>

<address>

<street>123 Main Street</street>

<city>Chennai</city>

<state>Tamil Nadu</state>

<zip>600001</zip>

</address>

<course>B.Sc Computer Science</course>

<hobbies>Reading, Coding, Music</hobbies>

</student>

<student>

<id>2</id>

<name>Jane Smith</name>

<dob>2001-08-15</dob>

<gender>Female</gender>

<email>jane@example.com</email>

<phone>9876543210</phone>

<address>

<street>45 Park Avenue</street>

<city>Delhi</city>

<state>Delhi</state>

<zip>110001</zip>

</address>

<course>B.Com</course>

<hobbies>Drawing, Swimming</hobbies>

</student>

</students>

readXML.js

```
const fs = require("fs");
```

```
const xml2js = require("xml2js");
```

```
// Read XML file
```

```
const xmlData = fs.readFileSync("stu.xml", "utf-8");
```

```
// Parse XML
```

```
const parser = new xml2js.Parser();

parser.parseString(xmlData, (err, result) => {

    if (err) throw err;

    const students = result.students.student;

    students.forEach(s => {

        console.log("Name:", s.name[0]);

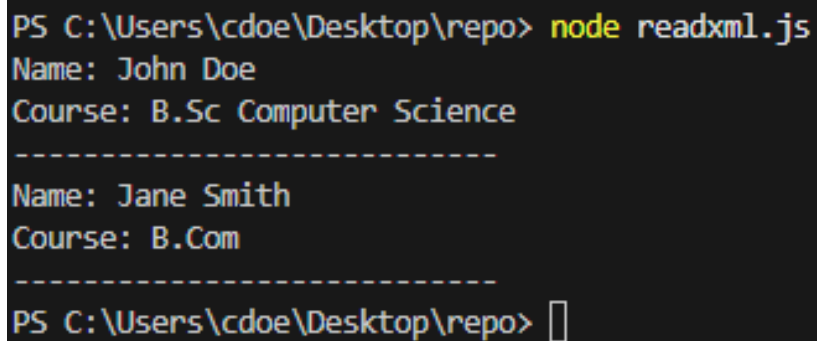
        console.log("Course:", s.course[0]);

        console.log("-----");

    });

});
```

Output:

A screenshot of a Windows PowerShell terminal window. The prompt is 'PS C:\Users\cdoe\Desktop\repo>'. The command 'node readxml.js' has been executed. The output is displayed in three lines: 'Name: John Doe', 'Course: B.Sc Computer Science', and a dashed line '-----'. This is followed by 'Name: Jane Smith', 'Course: B.Com', and another dashed line '-----'. The prompt 'PS C:\Users\cdoe\Desktop\repo>' is shown again at the bottom, followed by a cursor icon.

```
PS C:\Users\cdoe\Desktop\repo> node readxml.js
Name: John Doe
Course: B.Sc Computer Science
-----
Name: Jane Smith
Course: B.Com
-----
PS C:\Users\cdoe\Desktop\repo> █
```

Result:

The above program has been executed and verified successfully.